

# Recyclean: Plastic Waste Reward App

## 1.0 Introduction

RecyClean is a mobile and web-based system that is designed to encourage proper recycling of plastic bottles through a reward-based approach at the barangay level. Residents can submit plastics and earn points, staff can verify submissions using the mobile app, and administrators can manage the system through a web dashboard.

This user manual provides simple instructions for Residents, Staff, and Administrators on how to use the RecyClean system correctly and securely.

## 2.0 System Access and User Roles

| User Role      | System Access      | Main Responsibilities                                                                                                                                                                                                 |
|----------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resident User  | Mobile Application | <ul style="list-style-type: none"><li>• Scan QR codes to record plastic submissions</li><li>• Earn points and redeem rewards</li></ul>                                                                                |
| Drop-off Staff | Mobile Application | <ul style="list-style-type: none"><li>• Validate plastic submissions (AI or manual)</li><li>• Generate QR codes</li><li>• View submission analytics</li></ul>                                                         |
| Administrator  | Web Application    | <ul style="list-style-type: none"><li>• Manage the entire system</li><li>• View and export analytics</li><li>• Set point values and rewards</li><li>• Approve staff accounts</li><li>• Manage DIY tutorials</li></ul> |

### 3.0 Resident User Guide (Mobile Application)

This section provides step-by-step instructions for Resident Users on how to use the RecyClean mobile application to scan QR codes, earn points, redeem rewards, and view activity history.

#### 3.1 Resident Registration

New users must register an account before they can log in to the RecyClean mobile application.

##### Steps:

1. Open the RecyClean mobile application on your Android device.
2. Tap the Register button on the login screen.
3. Enter the required information (such as name, contact number, and password).

9:14 PM

<

### Create Account

Sign up to start earning rewards for recycling.

Username

3-20 characters • letters, numbers, and \_ only

First name Last name

Email

Mobile Number (Optional)

Password

- ☐ At least 8 characters
- ☐ Contains uppercase
- ☐ Contains lowercase
- ☐ Contains number
- ☐ Contains symbol (!@#\$%)

✔ Passwords match

Very weak

Confirm Password

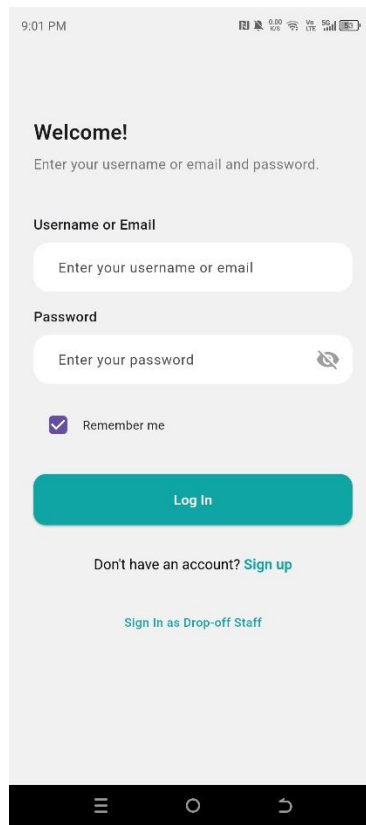
4. Review the information entered and tap Submit to create an account.

### 3.2 Resident Login

This feature allows registered residents to securely access the RecyClean mobile application.

#### Steps:

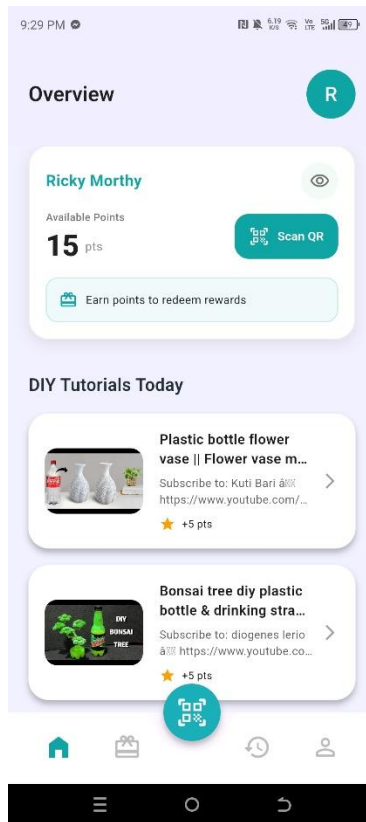
1. Open the RecyClean mobile application on your Android device.



2. Enter your registered username and password.
3. Tap the Login button to access the application.

### 3.2 Resident Dashboard

The dashboard serves as the main screen of the application, where residents can view their points and access key features.



#### Dashboard Features:

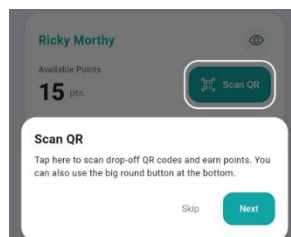
- Current points balance
- QR code scanner button
- DIY tutorials section
- Navigation menu

### 3.3 Scanning QR Code for Plastic Submission

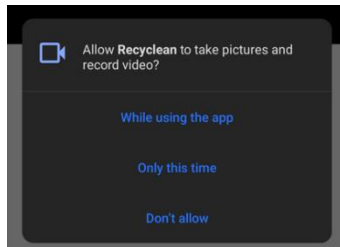
Residents scan a QR code provided by the drop-off staff after a plastic submission has been verified.

#### Steps:

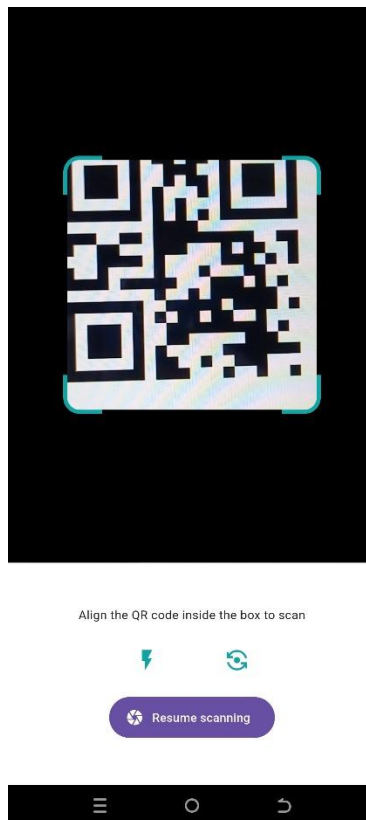
1. From the dashboard, tap the Scan QR Code button.



2. Allow camera access if prompted.



3. Point the camera at the QR code generated by the staff.

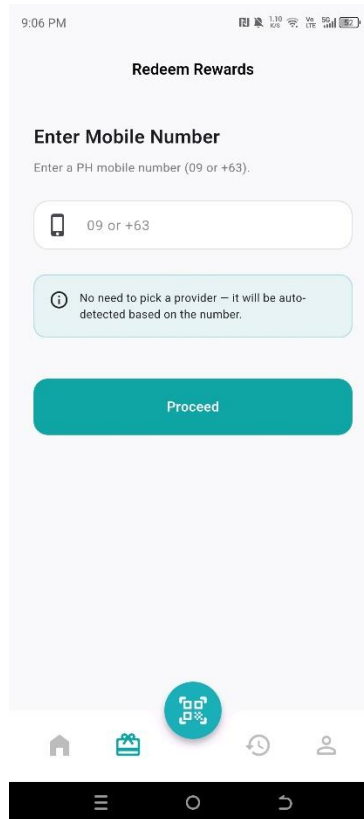


4. Wait for the confirmation message indicating that points have been credited.

### 3.4 Redeeming Rewards

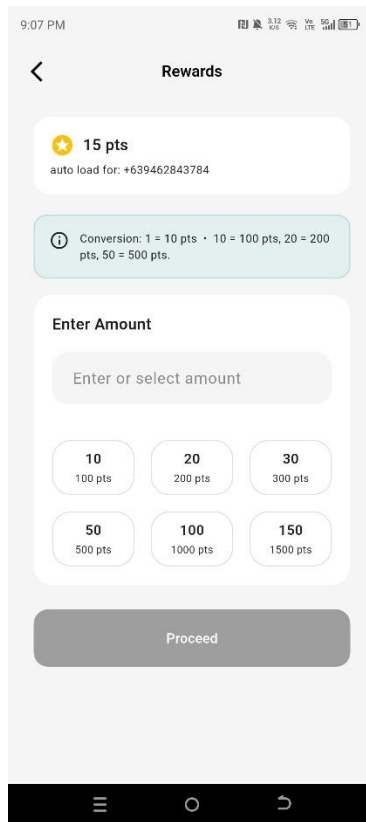
Residents can redeem earned points for available rewards such as mobile load.

#### Steps:



The screenshot shows a mobile application interface for redeeming rewards. At the top, the status bar displays the time as 9:06 PM and various system icons. The app's title bar reads "Redeem Rewards". Below this, the section is titled "Enter Mobile Number" with a subtitle "Enter a PH mobile number (09 or +63)". A text input field contains the placeholder "09 or +63" and a mobile phone icon. A light blue information box below the field states: "No need to pick a provider — it will be auto-detected based on the number." A large teal "Proceed" button is positioned below the information box. At the bottom of the screen, there is a navigation bar with four icons: a home icon, a shopping bag icon, a circular icon with a QR code (which is highlighted with a red circle), and a user profile icon. The very bottom of the image shows the standard Android navigation bar with back, home, and recent apps buttons.

1. Navigate to the Rewards section from the menu.
2. Select the desired reward amount.



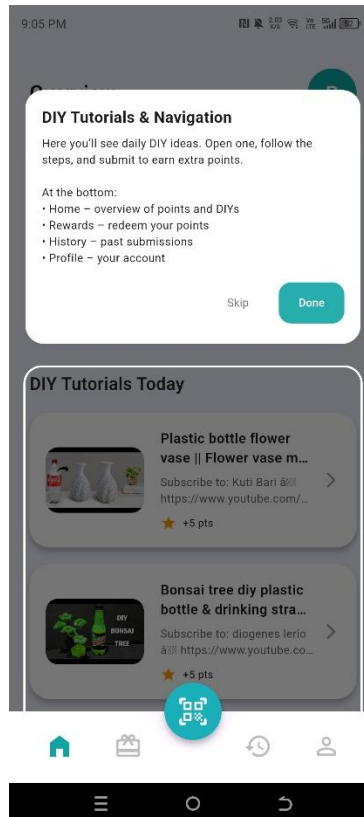
3. Tap Redeem and confirm the transaction.

### 3.5 DIY Recycling Tutorials

The DIY Tutorials feature provides educational content on creative recycling while allowing users to earn additional points.

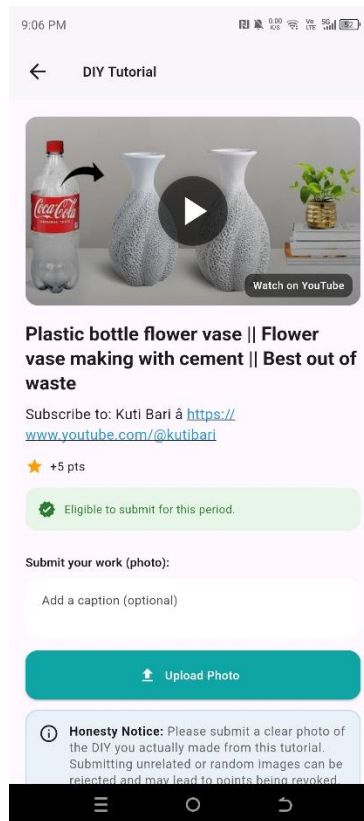
#### Steps:

1. Open the DIY Tutorials section from the dashboard or menu.





2. Select a tutorial from the list.



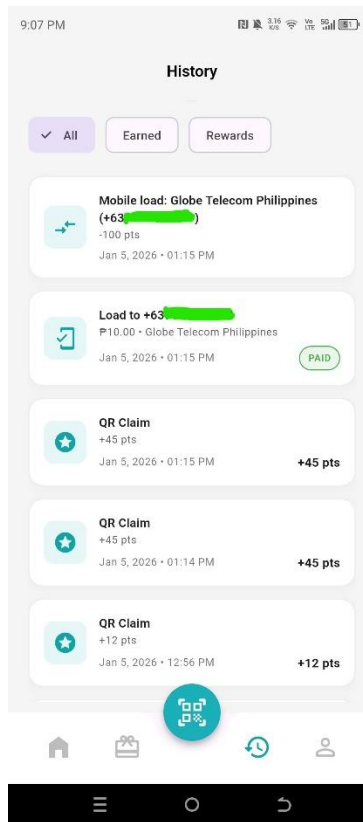
3. Follow the instructions provided in the tutorial.
4. Complete the activity and submit a photo of your work to earn bonus points.

### 3.6 Viewing Submission and Reward History

Residents can view their past plastic submissions and reward transactions for tracking and transparency.

#### Steps:

1. Navigate to the History section.
2. View the list of previous submissions and redeemed rewards.



3. Check dates, points earned, and transaction status.

#### 4.0 Drop-off Staff User Guide (Mobile Application)

This section provides instructions for Drop-off Staff on how to register, log in, validate plastic submissions, generate QR codes, and view submission analytics using the RecyClean mobile application.

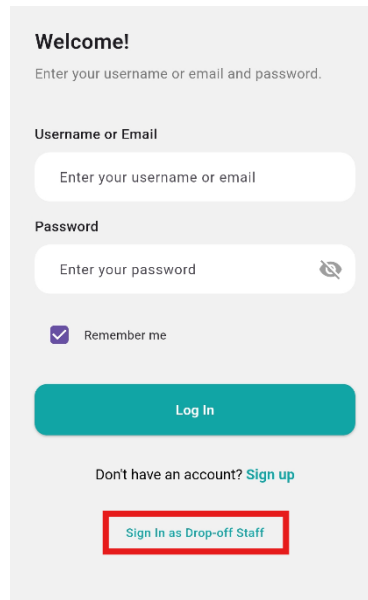
##### 4.1 Staff Registration

New staff members must register an account before they can access the system. Staff accounts require administrator approval before login is allowed.

##### Steps:

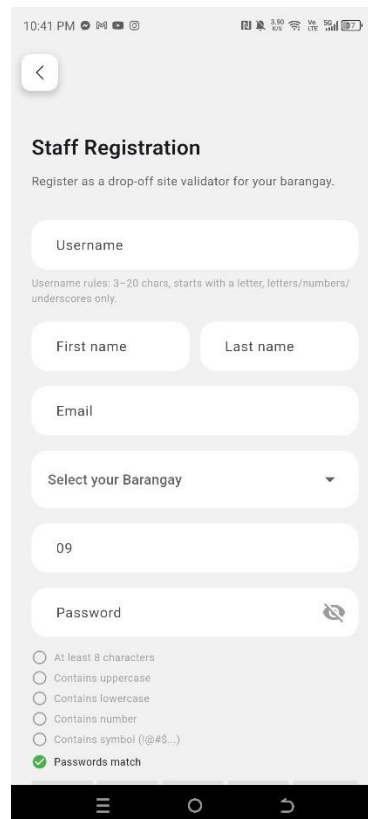
1. Open the RecyClean mobile application on the device provided.

2. Tap the Register as a Staff option on the login screen.



The image shows a login screen titled "Welcome!". Below the title is a subtitle "Enter your username or email and password." There are two input fields: "Username or Email" and "Password". Below the password field is a "Remember me" checkbox which is checked. A teal "Log In" button is present. At the bottom, there is a link "Don't have an account? Sign up" and a button "Sign In as Drop-off Staff" which is highlighted with a red rectangle.

3. Enter the required information (such as name, contact details, and assigned drop-off location).



The image shows a "Staff Registration" form. At the top, there is a subtitle "Register as a drop-off site validator for your barangay." The form includes several input fields: "Username", "First name", "Last name", "Email", "Select your Barangay" (a dropdown menu), and a phone number field with "09" entered. There is a "Password" field with a toggle icon. Below the password field, there are five radio button options for password requirements: "At least 8 characters", "Contains uppercase", "Contains lowercase", "Contains number", and "Contains symbol (!@#\$...)". The "Passwords match" option is checked with a green checkmark. The form is displayed on a mobile device screen with a status bar at the top showing the time as 10:41 PM and various icons.

4. Tap Submit Registration.

**Important:**

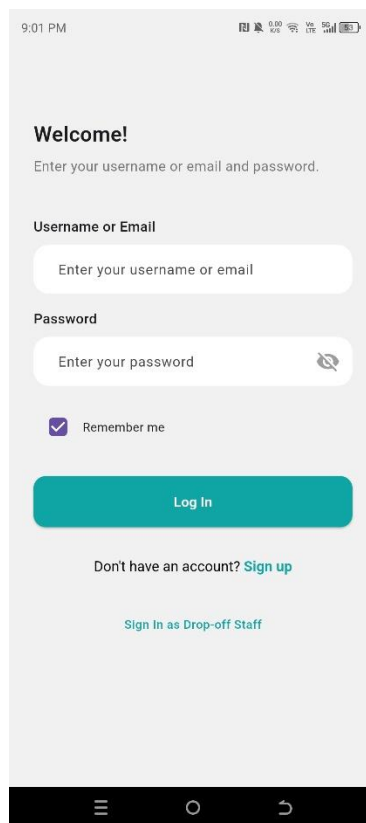
- After registration, the account will remain pending approval.
- Staff members cannot log in until the administrator approves the account.

## 4.2 Staff Login

Once the administrator approves the staff account, the staff member can log in to the system.

Steps:

1. Enter the approved staff username and password.
2. Tap the Login button to access the staff dashboard.



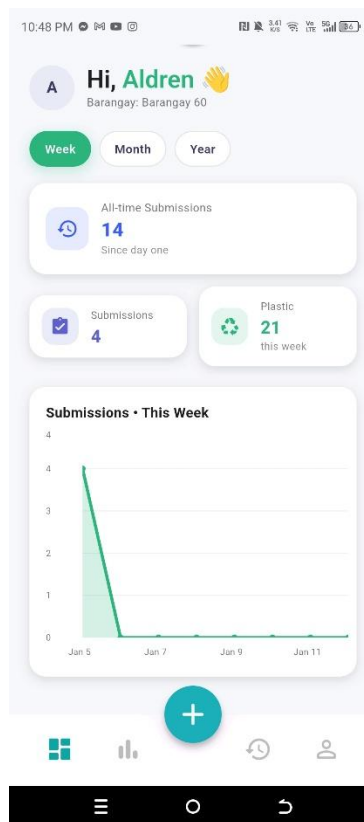
The screenshot shows a mobile application interface for staff login. At the top, the status bar displays the time as 9:01 PM and various system icons. The main content area has a light gray background. It starts with a 'Welcome!' heading followed by the instruction 'Enter your username or email and password.' Below this are two input fields: 'Username or Email' and 'Password', each with a placeholder text 'Enter your username or email' and 'Enter your password' respectively. A 'Remember me' checkbox is checked. A prominent teal 'Log In' button is centered below the inputs. At the bottom, there are two links: 'Don't have an account? Sign up' and 'Sign In as Drop-off Staff'. The bottom of the screen shows a black navigation bar with standard Android icons.

### 4.3 Staff Dashboard Overview

The staff dashboard is the main workspace where staff can manage plastic submissions and view analytics.

#### Dashboard Features:

- Create and validate plastic submissions
- Generate QR codes for verified submissions
- View personal submission analytics
- Access navigation menu

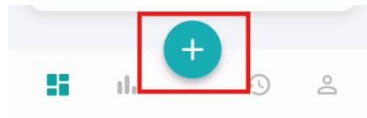


### 4.3 Creating a Plastic Submission (AI-Assisted or Manual)

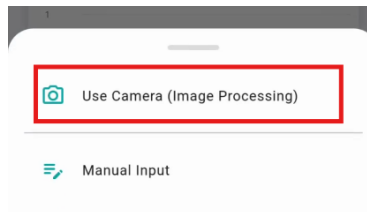
Staff members validate plastic bottle submissions brought by residents. Validation can be done using computer vision (AI-assisted) or manual entry.

### AI-Assisted Submission Steps:

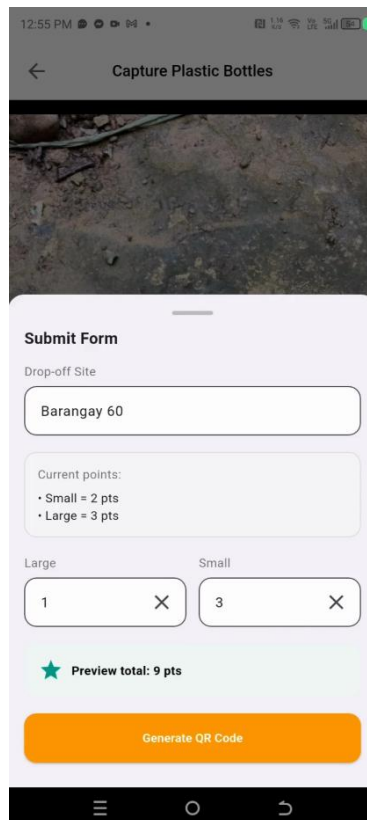
1. From the dashboard, tap Create Submission.



2. Select the Camera / AI-Assisted option.



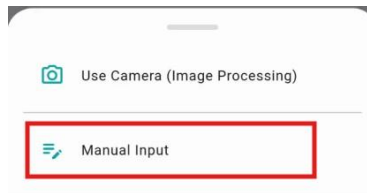
3. Capture an image of the plastic bottles using the device camera.
4. Review the AI-detected bottle count and size classification.



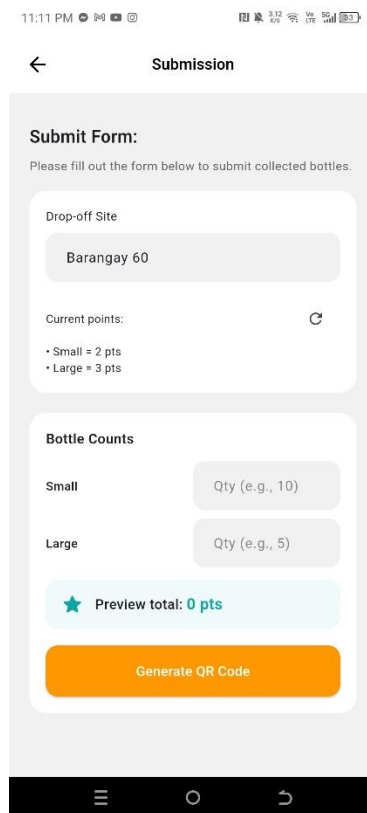
5. Adjust the values manually if necessary.
6. Tap Confirm Submission to proceed.

### Manual Submission Steps (if needed):

1. Select the Manual Entry option.



2. Enter the number and size of plastic bottles manually.

A screenshot of a mobile application interface showing a 'Submission' form. The form is titled 'Submit Form:' and includes instructions: 'Please fill out the form below to submit collected bottles.' The form has two main sections. The first section is 'Drop-off Site' with a text input field containing 'Barangay 60'. Below this is a 'Current points:' section with a refresh icon and a list of points: '• Small = 2 pts' and '• Large = 3 pts'. The second section is 'Bottle Counts' with two rows: 'Small' and 'Large'. Each row has a text input field for quantity, with 'Qty (e.g., 10)' and 'Qty (e.g., 5)' respectively. Below the input fields is a 'Preview total: 0 pts' section with a star icon. At the bottom of the form is a large orange button labeled 'Generate QR Code'. The status bar at the top shows the time as 11:11 PM and various icons.

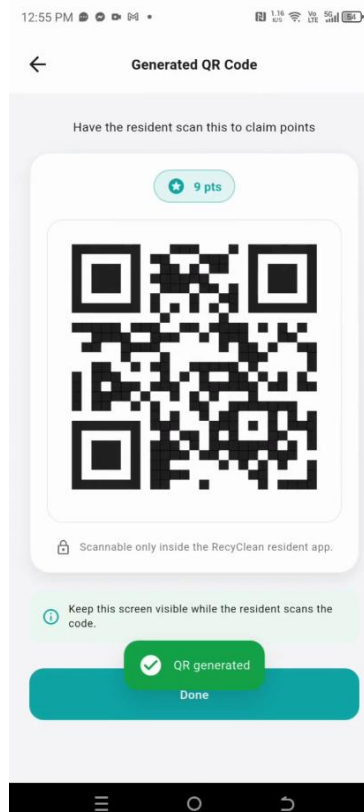
3. Review the entered data.
4. Tap Confirm Submission to proceed.

### 4.4 QR Code Generation

After a plastic submission is confirmed, the system automatically generates a QR code for the resident.

**Steps:**

- Complete and confirm the plastic submission.
- Wait for the system to generate the QR code.
- Display the QR code to the resident for scanning.

**Important Notes:**

- QR codes are unique per submission.
- QR codes do not contain personal information.
- Each QR code can only be scanned once.

**4.5 Viewing Submission Analytics**

Staff members can monitor their own submission activity and collected plastic bottle data using the analytics feature.

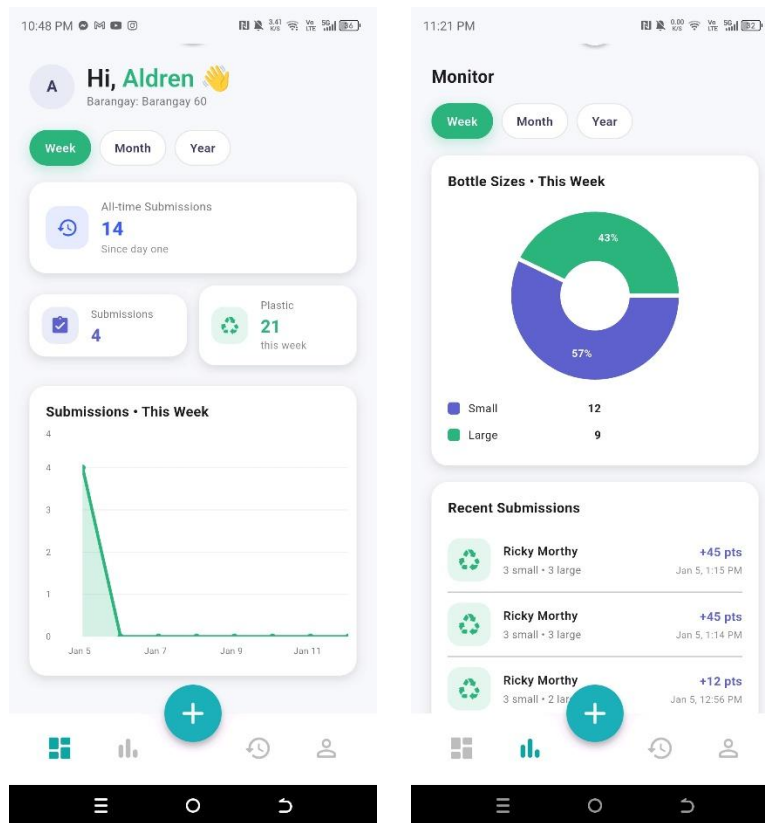


### Analytics Information Available:

- Total submissions made
- Bottles collected per week, month, and year
- Visual charts and summary data

### Steps:

1. Navigate to the Analytics section from the dashboard or monitor.



2. Select the desired time range (weekly, monthly, or yearly).
3. Review the displayed graphs and statistics.

## 5.0 Administrator User Guide (Web Application)

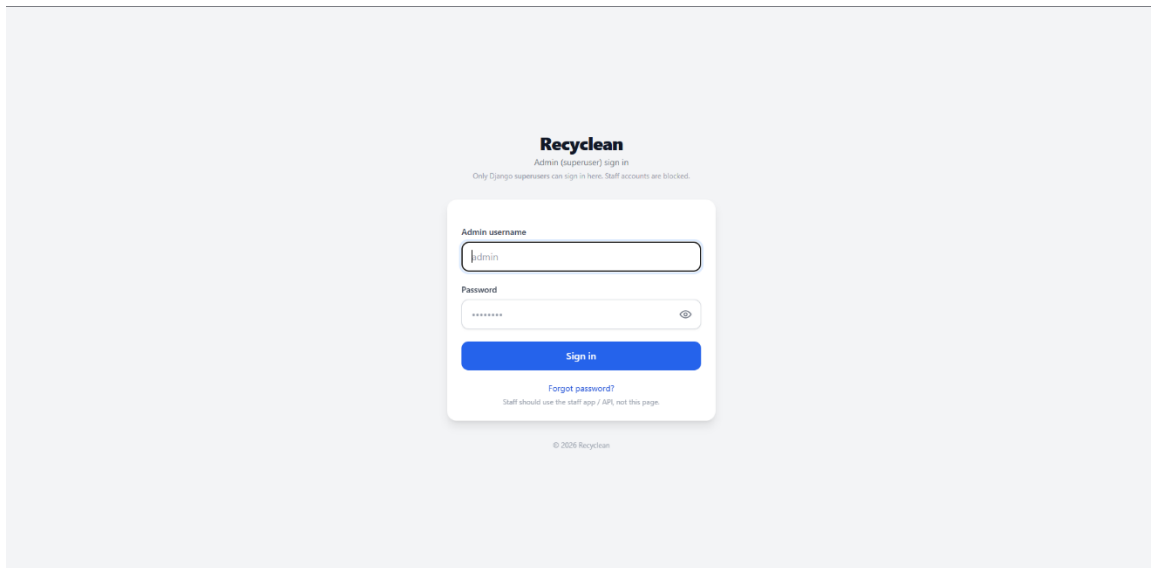
This section provides instructions for System Administrators on how to manage users, monitor plastic submissions, configure reward values, and maintain system content using the RecyClean web application.

## 5.1 Admin Login

Administrators access the RecyClean system through a secure web-based login page.

Steps:

1. Open a supported web browser (e.g., Google Chrome or Microsoft Edge).
2. Enter the RecyClean admin web application URL. <https://admin.recyclean.app/>
3. Input the assigned administrator username and password.
4. Click the Login button to access the admin dashboard.



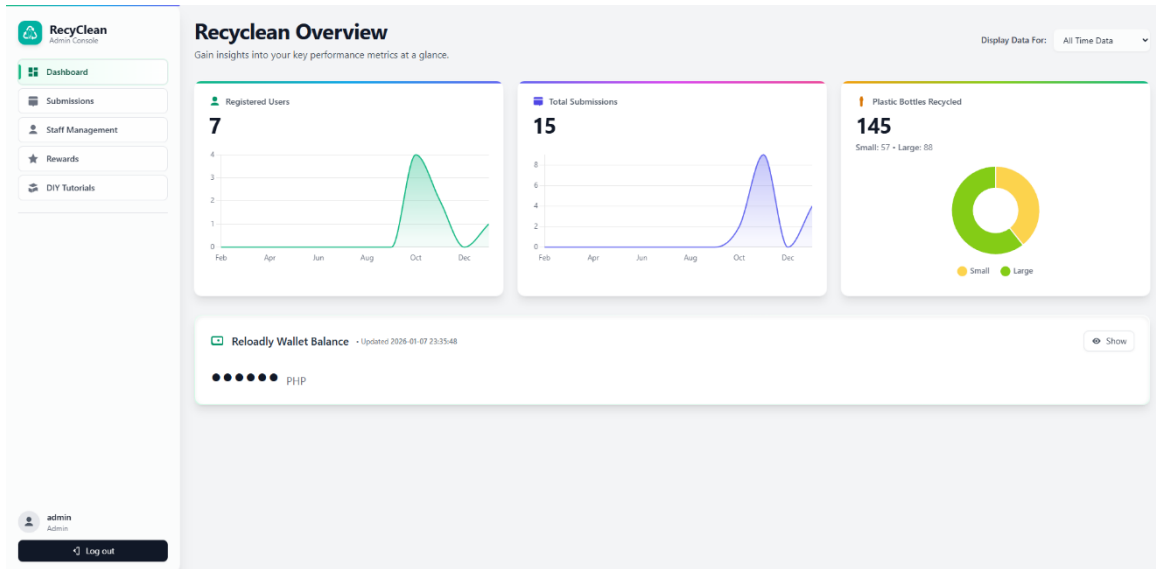
## 5.2 Admin Dashboard Overview

The admin dashboard provides a centralized view of system activity and performance.

Dashboard Features:

- Overview of total users and staff
- Total plastic submissions across all barangays
- Summary of reward transactions

- Visual charts and analytics



### 5.3 Submission Monitoring and Analytics

Administrators can monitor all plastic submissions recorded in the system.

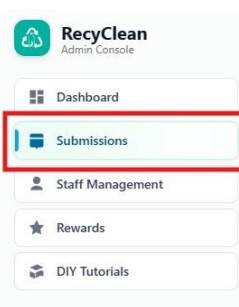
#### 5.3.1 Overall Submissions Monitoring

From the Submissions page, administrators can:

- View all registered drop-off sites (barangays)
- See assigned staff per barangay
- Review current point configuration per bottle size
- Export submission summaries across all drop-off sites

Steps:

1. Navigate to the Submissions section from the sidebar menu.



2. Review the list of drop-off site and assigned staff.
3. Select a date range if needed.

From  To  [Export CSV](#)

4. Click Export All (CSV) to download submission data from all drop-off sites

The screenshot displays the 'Submissions' page in the RecyClean Admin Console. The left sidebar shows the navigation menu with 'Submissions' selected. The main content area has a header 'Submissions' with a subtitle 'Manage points configuration, review drop-off sites, and export summaries.' and a date range selector. Below this, there are two main sections: 'Points per Bottle' and 'Drop-off Sites'. The 'Points per Bottle' section shows a table with columns for Small bottle points, Large bottle points, and Points per P1, with values 2, 3, and 10 respectively. The 'Drop-off Sites' section shows a table with columns for BARANGAY, STAFF, EXPORT RANGE, and ACTIONS. Two rows are visible: 'Barangay 60' with staff 'Aldren Reachord' and 'Barangay 63' with staff 'John Due' and 'Stephanie Lourdes'. Both rows have 'View' and 'Export CSV' buttons.

### 5.3.2 Points Configurations

Administrators may edit the points allocated per bottle size and the value of points in peso.

Steps:

1. From the Submission page, click the Unlock Editing in Points per Bottle section.



## 2. Verify by entering admin password

Points per Bottle Unlock editing

Enter your **admin password** to enable editing for 5 minutes.

Admin password Verify Cancel

| Small bottle points | Large bottle points | Points per P1 |
|---------------------|---------------------|---------------|
| 2                   | 3                   | 10            |

Save Last updated: 2026-01-05 13:22

3. Now you'll be able to edit for 5 minutes, once 5 minutes is over it will be lock again

4. Once done editing click save and lock it.

Points per Bottle Editing unlocked Lock now

| Small bottle points | Large bottle points | Points per P1 |
|---------------------|---------------------|---------------|
| 2                   | 3                   | 10            |

Save Last updated: 2026-01-05 13:22

Editing unlocked for 4:54. Make your changes and click Save.

### 5.3.3 Viewing Submissions Per Barangay

Administrators may view detailed submission data for a specific barangay by selecting the View option.

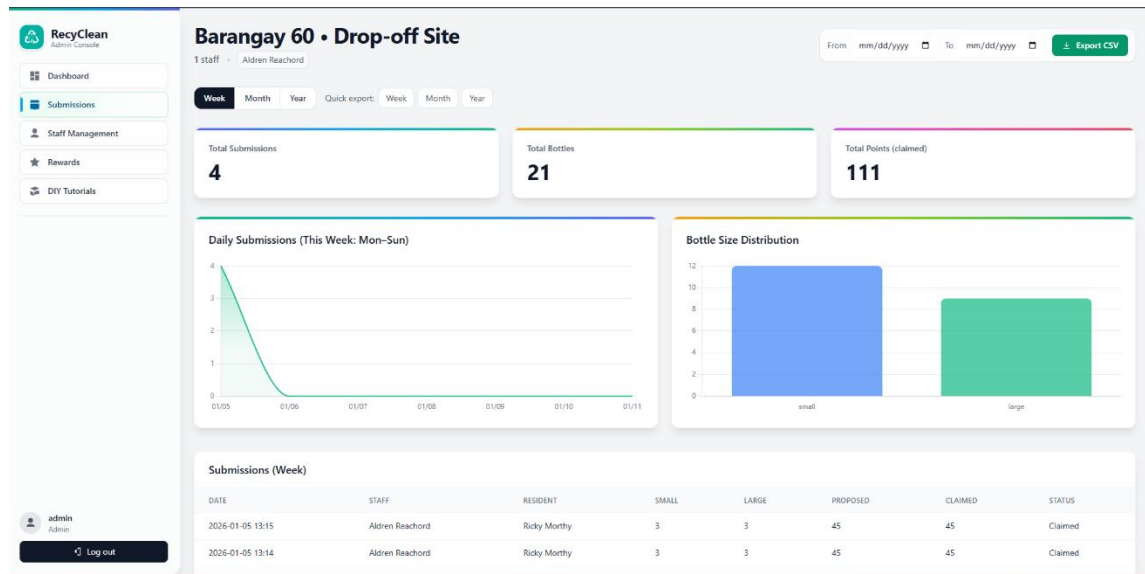
#### Available Information Per Barangay:

- Total submissions within the selected period
- Total bottles collected
- Total points claimed
- Daily submission trends
- Bottle size distribution (small and large)
- Individual submission records (per staff and resident)

#### Steps:

1. From the Submissions page, click View beside a barangay.
2. Select a time range (Week, Month, or Year).

### 3. Review summary cards, charts, and submission tables.

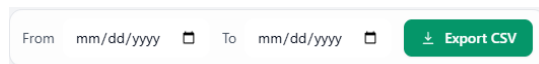


#### 5.3.4 Exporting Barangay Submission Data

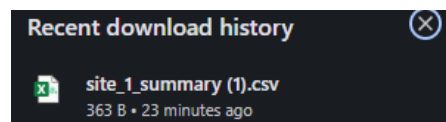
Administrators can export submission data specific to a barangay for documentation or reporting purposes.

##### Steps:

1. While viewing a barangay's submission details, select a date range if needed.
2. Click Export CSV.



3. Download the generated CSV file containing submission records for the selected barangay only.



##### Important Notes:

- Export All (CSV) downloads combined data from all drop-off sites.
- Export CSV (Barangay View) downloads data for the selected barangay only.

- Exported files include submission dates, staff, bottle counts, and point values.

## Purpose of This Feature

This feature supports transparency, monitoring, and reporting by allowing administrators to analyze recycling activity at both the system-wide and barangay-specific levels.

## 5.4 Staff Account Management

Administrators are responsible for approving and managing staff accounts before system access is granted.

### Steps:

1. Navigate to the Staff Management section from the dashboard menu.
2. Review the list of pending staff registration requests.
3. Click Approve to allow staff access, or Reject to deny the request.
4. Manage existing staff accounts as needed.

**RecyClean**  
Admin Console

- Dashboard
- Submissions
- Staff Management**
- Rewards
- DIY Tutorials

**Staff Management**  
Review pending requests, manage active/inactive staff, and inspect transactions.

**Pending Staff Requests**

| NAME              | BARANGAY | MOBILE | ACTIONS |
|-------------------|----------|--------|---------|
| No pending staff. |          |        |         |

**Active Staff**

| NAME              | BARANGAY    | MOBILE        | ACTIONS                                                      |
|-------------------|-------------|---------------|--------------------------------------------------------------|
| Aldren Reachord   | Barangay 60 | +639123456789 | <a href="#">View Transactions</a> <a href="#">Deactivate</a> |
| Stephanie Lourdes | Barangay 63 | +639543237463 | <a href="#">View Transactions</a> <a href="#">Deactivate</a> |
| John Due          | Barangay 63 | +639289607276 | <a href="#">View Transactions</a> <a href="#">Deactivate</a> |

**Inactive Staff**

| NAME               | BARANGAY | MOBILE | ACTIONS |
|--------------------|----------|--------|---------|
| No inactive staff. |          |        |         |

admin  
Admin  
[Log out](#)

## 5.6 Reward Transaction Monitoring and Export

Administrators can monitor all reward redemption transactions and export system data.

## Steps:

- Navigate to the Reward Transactions section.
- View the list of completed and pending redemptions.

The screenshot displays the RecyClean Admin Dashboard. On the left is a sidebar with navigation links: Dashboard, Submissions, Staff Management, Rewards (highlighted), and DIY Tutorials. The main content area is titled 'Reloadly Overview' with the subtitle 'Monitor wallet balance, redemption counts, and recent transactions.' It features three summary cards: 'Reloadly Wallet Balance' showing a PHP balance, 'Reward Redeemed' showing a count of 6, and 'Recent Activity' with a description. Below these are two tables. The 'Reloadly Transactions' table lists transactions with columns for TXN ID, DATE, OPERATOR, PHONE, AMOUNT, and STATUS. The 'Reward Transaction Log (App)' table lists requests with columns for USER, MOBILE & TELCO, AMOUNT, STATUS, DATE REQUESTED, RELOADLY REF, and NOTE. The user 'admin' is logged in, and a 'Log out' button is visible at the bottom left.

| TXN ID   | DATE                | OPERATOR                  | PHONE        | AMOUNT | STATUS   |
|----------|---------------------|---------------------------|--------------|--------|----------|
| 28315436 | 2025-11-01 09:08:53 | Smart Philippines         | 639289607276 | 10 PHP | Approved |
| 28316124 | 2025-11-01 10:56:03 | Globe Telecom Philippines | 639063995973 | 10 PHP | Approved |
| 28354849 | 2025-11-04 18:44:43 | Globe Telecom Philippines | 639051276921 | 10 PHP | Approved |
| 28369988 | 2025-11-06 02:15:04 | DITO Philippines          | 639934909468 | 10 PHP | Approved |
| 29280886 | 2026-01-05 05:15:41 | Globe Telecom Philippines | 639051276921 | 10 PHP | Approved |

| USER | MOBILE & TELCO         | AMOUNT     | STATUS | DATE REQUESTED     | RELOADLY REF | NOTE |
|------|------------------------|------------|--------|--------------------|--------------|------|
|      | +639051276921<br>globe | ₱10.00 PHP | Paid   | Jan 05, 2026 13:15 | -            | ---  |
|      | +639934909468<br>dito  | ₱10.00 PHP | Paid   | Nov 06, 2025 10:15 | -            | ---  |

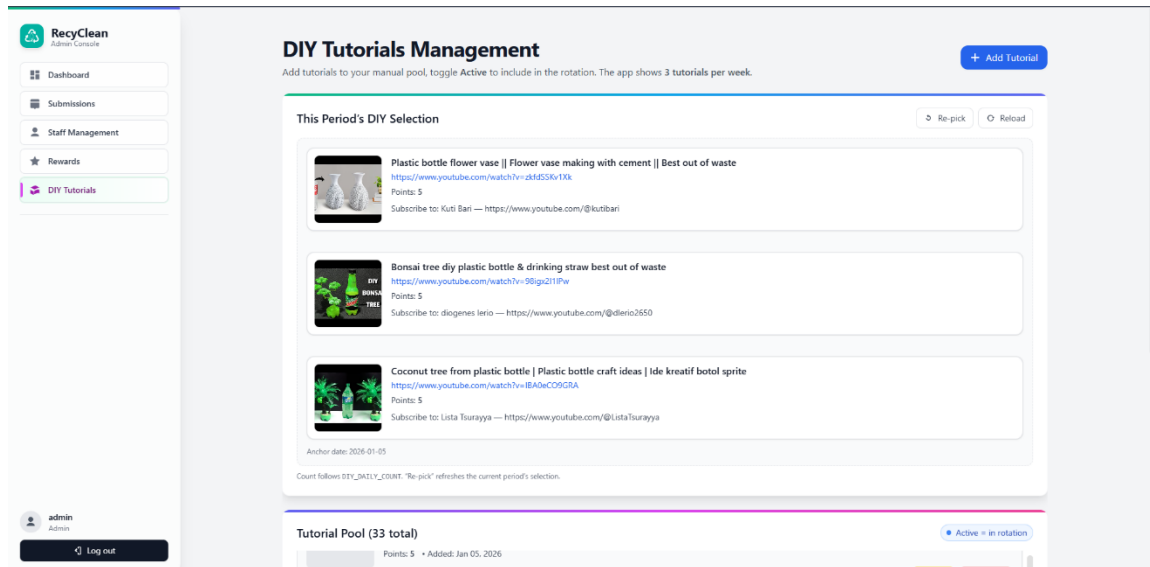
## 5.7 DIY Tutorial Management

Administrators manage the pool of DIY recycling tutorials available to users.

### Responsibilities:

- Add new DIY tutorial videos
- Assign point values to tutorials
- Enable or disable tutorials



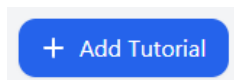


### 5.7.1 Adding a DIY Tutorial

Administrator manages the DIY Tutorials that are showed in the mobile application.

Steps:

1. Navigate to the DIY Tutorials management section.
2. Click Add Tutorial.



3. Copy the URL link of the Youtube video and paste.

**Add Tutorial**

Title Points on Submit

0

Instructions / Description

Steps, tips... (you can use -, \*, + for bullets)

Tip: Start lines with "-", "\*", or "+" for bullet points.

**Video URL (YouTube)** Duration (seconds)

☒ Auto-fill from YouTube ☐ Refresh metadata from YouTube

Fetch title, duration, and thumbnail/description from the video URL even if fields are already filled.

Thumbnail Status

Choose File No file chosen ☒ Active in Pool

Leave empty to keep current image when editing.

**Save Tutorial**

4. Click the Autofill to fill the info of the Youtube video

Video URL (YouTube) Duration (seconds)

☒ Auto-fill from YouTube ☐ Refresh metadata from YouTube

Fetch title, duration, and thumbnail/description from the video URL even if fields are already filled.

5. Assign points on submission.

**Points on Submit**

2

6. Click Save to add the tutorial in the DIY pool.

### 5.7.2 Managing the Tutorial Pool

The Tutorial Pool contains all available DIY tutorials stored in the system.

#### Available Actions:

- Edit – Modify tutorial details such as title, link, or point value

- Delete – Remove a tutorial from the pool
- Active / Inactive Toggle – Control whether a tutorial is included in the rotation shown to users

### 5.7.3 This Period's DIY Selection

The system automatically selects a limited number of tutorials from the active pool to be displayed in the mobile application for a given period.

#### Administrator Controls:

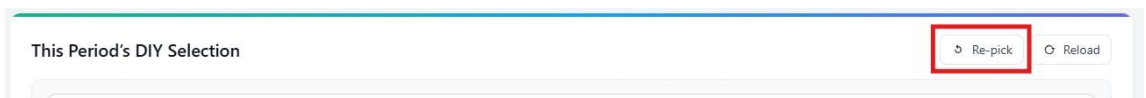
- Re-pick – Randomly selects a new set of tutorials from the active pool
- Reload – Refreshes the current selection without changing the chosen tutorials

#### Important Notes:

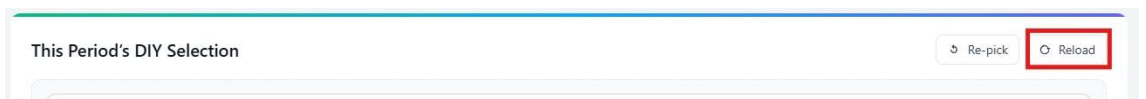
- Only active tutorials are eligible for selection
- The number of tutorials displayed per period is system-defined
- Changes made by the administrator are reflected in the mobile app after refresh

#### Steps to Manage DIY Tutorials

1. Navigate to the DIY Tutorials section from the admin sidebar menu.
2. Review the current tutorial pool and active tutorials.
3. Add, edit, or delete tutorials as needed.
4. Use Re-pick to change the tutorials displayed in the mobile app.



5. Use Reload to refresh the current tutorial list.



## 6.0 Security and Data Privacy Guidelines

The RecyClean system is designed with security and data privacy considerations to ensure safe and responsible system use.

### Security Measures

- Role-based access control is implemented to restrict system functions based on user roles (Resident, Staff, Administrator).
- Only approved staff accounts are allowed to access staff-specific features.
- Administrative actions are limited to authorized administrator accounts.

### Data Privacy Measures

- QR codes generated by the system do not contain personal information.
- User data collected by the system is limited to what is necessary for system operation.
- All user and transaction data are securely stored and processed within the system.

### Legal Compliance

- The system is designed to comply with the Data Privacy Act of 2012 (Republic Act No. 10173).
- Personal information is handled responsibly and used only for system-related purposes.

## 7.0 Troubleshooting

This section provides solutions to common issues encountered while using the RecyClean system.

| Issue                              | Possible Solution                                                      |
|------------------------------------|------------------------------------------------------------------------|
| Unable to log in                   | Check internet connection and ensure correct username and password     |
| Staff account cannot log in        | Ensure the staff account has been approved by the administrator        |
| QR code not scanning               | Ensure camera permission is enabled and the QR code is clearly visible |
| Points not credited after scanning | Refresh the application or check submission status                     |

|                                 |                                                       |
|---------------------------------|-------------------------------------------------------|
| Reward redemption not reflected | Verify available points and check transaction history |
| System not loading properly     | Restart the application or refresh the web page       |

## 8.0 Summary

The RecyClean System User Manual provides clear instructions for residents, staff, and administrators on how to use the system effectively. By following the procedures outlined in this manual, users can ensure proper plastic submission, accurate reward processing, and secure system operation. This guide supports consistent system use, transparency, and responsible data handling across the RecyClean platform.